

kolum: competitive read, and what it means for our CBAM calculator

Internal brief · 3 August 2026 · prepared for the founding team

Update — 8 August 2026

Two of the four recommendations below are now done or substantially done, on a branch that shipped multi-line estimates, a real annual threshold verdict, and export. Full detail: `docs/cbam-release-2026-08.md`. The analysis beneath this section is otherwise untouched from 3 August and still holds — including the Column A/B worked example below, checked again today against the live pack.

- **Recommendation 1, multi-line input — done.** The calculator now takes a whole bill of materials: add a line, remove a line, attest a calendar year, export the set. The "Where they are ahead" entry for this is corrected below, in place and dated, rather than removed.
 - **Recommendation 4, CSCF legibility — substantially done, not fully.** The printed export's fourth section and a corrected floor-not-ceiling statement on the calculator's own results card now explain the limit as a designed feature rather than a bare refusal. What has not moved: the on-screen tag is still the same amber "pending" it always was, and none of this language has reached outward-facing copy — marketing, sales conversations — where "the reason to trust us" actually has to land on a prospect.
 - **Recommendation 2, sourcing comparison — not built.** It was already the highest commercial-value item on this list on 3 August. With recommendation 1 done and recommendation 4 substantially done, it is now the largest gap remaining and should be read as the top recommendation.
 - **Recommendation 3, Column A unreachable — unchanged.** Still a hold; the reasoning was not revisited and does not need to be.
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Why this document

We are about to spend effort improving the CBAM calculator. Before doing that, it is worth knowing precisely what the best-funded competitor in this space has already built, where they stop, and which of those stopping points are deliberate.

The short version: **kolum is ahead of us on product surface and behind us on regulatory honesty.** Neither gap is accidental, and only one of them is worth closing.

What kolum is

A German trade-compliance platform selling three regulated products — EU CBAM, UK CBAM and EUDR — with PPWR, tariff classification and customs filings announced as coming. Customers named publicly include BMW Group, CNH Industrial, GlobalFoundries and Schmitz Cargobull. They hold ISO 27001, ISO 9001 and TÜV SÜD certification.

Their commercial engine is a free, ungated CBAM calculator that converts into a sales demo. That calculator is our direct comparison point, so most of this brief concerns it.

Seven named case studies sit on their site. Two are worth reading twice:

- **Keller & Kalmbach** — "replaced an underperforming CBAM tool with kolum." There is already displacement happening in this market.
- **Böllhoff** — "achieved over 70% of actual emissions data from suppliers." This is their headline proof metric, and it tells you what the product actually is: a supplier-data-collection machine with compliance reporting attached.

Their calculator, field by field

Field	Detail
Country of production	~180 options, non-EU only, includes UK
Import year	2026 / 2027 / 2028
Product → CN code	8-digit, type-ahead search
Import quantity (t)	numeric
Embedded emissions	optional — "leave empty to use official EU default values"
+ Add product	multi-line — several goods in one estimate

Their stated method, quoted from the page: emissions intensity (defaults per CN code and country, or the user's own actual values), multiplied by "the benchmark emissions level for that product category (used to determine how much is exempt)", by the CBAM factor of 97.5% for 2026, at the Q1 2026 certificate price of €75.36.

They cite EU Regulation 2025/2621 and 2025/2620 — **the same two instruments our own engine cites**. We are reading the same law.

Where we are genuinely ahead

1. The Column A / Column B distinction

This is the most consequential decision in a CBAM calculation and kolum's calculator does not expose it at all.

Free allocation is the deduction that keeps imports symmetric with EU producers who still receive free ETS allowances. It is subtracted from embedded emissions before certificates are counted, so an error here moves the invoice, not a report line. There are two benchmark columns:

- **Column B** — the whole finished good, precursors already embedded.
- **Column A** — this installation's own process step only, with precursors carried separately at their own Column B values.

The column is chosen by the *scope of the emissions figure being deducted from* — not by whether the data was verified. Our engine encodes this as a type-level constraint and refuses the inverse error outright.

Worked from our own rule pack, the cement chain shows why it matters:

CN	Good	Column A	Column B
25231000	Cement clinker (route A)	0.666	0.666
25232900	Other Portland cement	0	0.666
25232100	White Portland cement	0	0.859

A clinker kiln calcines limestone; the carbon comes out of the rock in that kiln, so its own process benchmark and its whole-product benchmark are identical. A cement grinder buys clinker and mills it — a mechanical operation emitting almost nothing — so its Column A is zero, and its entire free allocation arrives through the precursor term instead.

The arithmetic closes exactly: $0 \text{ (process)} + 1 \text{ t} \times 0.666 \text{ (clinker's own Column B)} = 0.666 = \text{Portland cement's Column B}$. **Column A plus the precursor equation is identical to Column B**, when done properly.

Get it wrong and you deduct zero where 0.666 was owed. On 100 t of CN 25232900 that is roughly 64.9 tCO_{2e}, about **€4,890**, silently removed from the importer's favour. Across our shipped pack, of the 269 (CN code, production route) pairs that publish both columns, 240 disagree — 89%. This is not an edge case.

Figure recounted 8 August 2026. The original read "102 of 141 — 72%", measured against the pack as it stood on 3 August; the re-vendor in [d529b1b](#) changed the benchmark table underneath it and it could not be reproduced. Counted here by (CN, route), which is the unit a benchmark is actually selected by — counting by CN code alone gives 549 of 570, 96%. The direction of the argument is unchanged and slightly stronger than when first written.

Our position: the full Column A path is implemented and audited, and deliberately unreachable from the public tool, because it is only correct against a verified process-only figure with a declared precursor list. A screening tool can obtain neither. That restraint is correct and we should keep it.

2. The CSCF, and what we refuse to fabricate

The cross-sectoral correction factor is unpublished for 2026–2030. It is **not** 1.0 by default. Our engine treats this as a first-class unknown: it returns a `cscf_pending` status carrying a clearly labelled what-if at the last value the Commission actually set, and never presents it as a finished figure.

kolum's published methodology lists four terms — intensity, benchmark, CBAM factor, certificate price — and the CSCF is not among them. Their calculator returns a single confident euro number.

A caveat we should state plainly: the string "CSCF" appears nowhere in their 2.98 MB front-end bundle, but I could not determine where their calculator sources its values, so I cannot rule out a correction applied server-side. What is certain is that **their published method does not disclose one**. That is a transparency fact, not a claim about their internals, and we should only ever make the former claim in public.

This is the sharper contrast, and it is a stance rather than a feature. Ours is the more honest instrument. Theirs is the more satisfying one. That tension is the real design problem in front of us.

Where they are ahead

Multi-line input — closed 8 August 2026. Their `+ Add product` lets an importer estimate a whole bill of materials. Ours handled one line when this was written on 3 August, and that was the single largest usability gap on this page. It is now closed — see `docs/cbam-release-2026-08.md` for what shipped. Kept here rather than deleted, as the dated record of what this brief found at the time.

Sourcing comparison. Their platform ranks every non-UK origin for a given CN code by CBAM-liable tonnage and surfaces the cheapest compliant option — they advertise "1,142 sourcing options". This is the most commercially valuable thing on their site, and **our rule pack already contains every input it needs**.

Limits stated as notes, not refusals. Their disclaimer reads: "While CBAM allows for deductions based on verified carbon pricing already paid in the country of origin, those mechanisms are not yet fully defined. For this reason, this calculator does not currently factor in foreign carbon costs." They name a known omission plainly and still ship a number. Our honesty rules are stricter but are expressed as refusals. There is something to learn in how they phrase a limit without withholding the answer.

UK CBAM: four facts worth banking

From their FAQ, and consistent with what we know of the UK mechanism:

1. UK CBAM is a **levy administered by HMRC**, not a certificate purchase-and-surrender market like the EU's. Certificate-trading features do not port across.
2. **Glass and ceramics are out** of initial 2027 scope.

3. **Indirect (electricity) emissions are delayed to 2029** at the earliest. Our estimator already hardcodes direct-only, which is exactly right for UK 2027.
4. Default values will exist so trade can continue.

One sign their UK work is early: their UK page quotes a certificate price of **£75.36/tCO₂e** while their EU calculator uses **€75.36**. Same numeral, different currency. Those UK figures look like carried-over placeholders rather than researched values.

Two structural observations

They are buying traffic their SEO is not earning. The site is a React single-page app with no `sitemap.xml` and no `robots.txt`; every path returns the same shell. Product pages are pre-rendered, but blog and case-study bodies are not, so their highest-value content is likely invisible to search engines. Consistent with this, the link that started this review was a paid Google Ads click on the keyword "uk cbam 2027".

Their moat is the supplier network, not the mathematics. Every module — emissions monitoring, article management, sourcing — resolves to the same asset: suppliers who have already been onboarded and are submitting actual data. The Böllhoff 70% figure is the product. The calculator is the top of a funnel that ends there.

Recommendation

1. **Add multi-line input.** Cheapest meaningful win, closes the most visible gap. **Done, 8 August 2026** — `docs/cbam-release-2026-08.md`.
 2. **Build the sourcing comparison.** Highest commercial value, and the data is already in our pack. This is the feature that makes a calculator a tool people return to. **Not built. Now the top recommendation** — the only one of the four still fully open, and it was already the highest commercial-value item here on 3 August.
 3. **Keep Column A unreachable.** Our audit's reasoning holds. Do not be tempted by feature parity into exposing a control that moves the bill against the user when entered wrongly. **Unchanged.** Still a hold.
 4. **Make the CSCF stance legible, not just correct.** Right now our honesty reads as a refusal. It should read as the reason to trust us. This is a copy and interface problem, not an engineering one, and it is the only thing on this list that competitors cannot copy by shipping a feature. **Substantially done, 8 August 2026** — the printed export's fourth section and a corrected floor-not-ceiling statement on the results card explain the limit as a designed feature now, not a bare refusal. Not fully closed: the card's own tag is still the same amber "pending," and the framing has not yet reached outward-facing copy, where it would have to persuade a prospect rather than a user already on the results page.
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Sources: *kolum.earth* product, calculator and case-study pages, retrieved 3 August 2026; our own rule pack `estimator-pack.json` (generated 29 July 2026) and `cbam-algos/cbam/sefa.ts`. Benchmark figures quoted are from our shipped pack, not from *kolum*.